

AQI Prediction System

Air Quality Index Prediction for Odisha, India
Using XGBoost Machine Learning

Comprehensive Project Report
Covering Dataset, EDA, Model Training, Evaluation & Deployment

Powered by Python, XGBoost, scikit-learn, FastAPI

1. Introduction

Air pollution is one of the most critical environmental challenges in India. The Air Quality Index (AQI) provides a standardized measure of air quality that helps the public understand how polluted the air is and what associated health effects might be of concern.

This project develops a Machine Learning-based AQI prediction system specifically designed for locations across Odisha, India. The system uses an XGBoost regression model trained on pollutant concentrations (PM2.5, PM10, NO2, SO2, CO, O3) and weather parameters (Temperature, Humidity) to predict AQI values and classify them into CPCB (Central Pollution Control Board) categories.

1.1 Objectives

- Generate a realistic synthetic AQI dataset for 400+ locations across 9 Odisha districts
- Perform comprehensive Exploratory Data Analysis (EDA)
- Train and evaluate XGBoost with hyperparameter tuning via GridSearchCV
- Build a RESTful API using FastAPI for real-time predictions
- Provide interactive CLI and batch CSV prediction modes

1.2 AQI Categories (Indian CPCB Standard)

AQI Range	Category	Health Impact
0 - 50	Good	Minimal impact
51 - 100	Satisfactory	Minor breathing discomfort (sensitive)
101 - 200	Moderate	Breathing discomfort (asthma, etc.)
201 - 300	Poor	Breathing discomfort on prolonged exposure
301 - 400	Very Poor	Respiratory illness on prolonged exposure
401 - 500	Severe	Health emergency, affects healthy people

2. Dataset Description

The dataset is synthetically generated using realistic parameter ranges derived from actual Odisha air quality monitoring data. It covers 9 districts with over 400 locations and spans 4 years (2022-2025) with 5,000+ data points.

2.1 Input Parameters

Parameter	Unit	Description	Source
PM2.5	ug/m3	Fine particulate matter (<2.5um)	Air Quality API
PM10	ug/m3	Coarse particulate matter (<10um)	Air Quality API
NO2	ppb	Nitrogen Dioxide	Air Quality API
SO2	ppb	Sulfur Dioxide	Air Quality API
CO	mg/m3	Carbon Monoxide	Air Quality API
O3	ppb	Ozone	Air Quality API
Temperature	C	Ambient temperature	Weather API
Humidity	%	Relative humidity	Weather API

2.2 Districts Covered

District	Locations	Key Areas
Kalahandi	109	Bhawanipatna, Kesinga, Dharmagarh...
Dhenkanal	181	Dhenkanal, Kamakhyanagar, Hindol...
Keonjhar	18	Keonjhar, Barbil, Joda, Champua...
Khordha	18	Bhubaneswar, Jatni, Khordha...
Jajpur	18	Jajpur, Vyasaganagar, Sukinda...
Cuttack	1	Cuttack
Sundargarh	1	Rourkela
Ganjam	1	Berhampur
Sambalpur	1	Sambalpur

3. Exploratory Data Analysis

Comprehensive EDA was performed to understand the distribution of AQI values, seasonal patterns, location-wise variations, and correlations between pollutants.

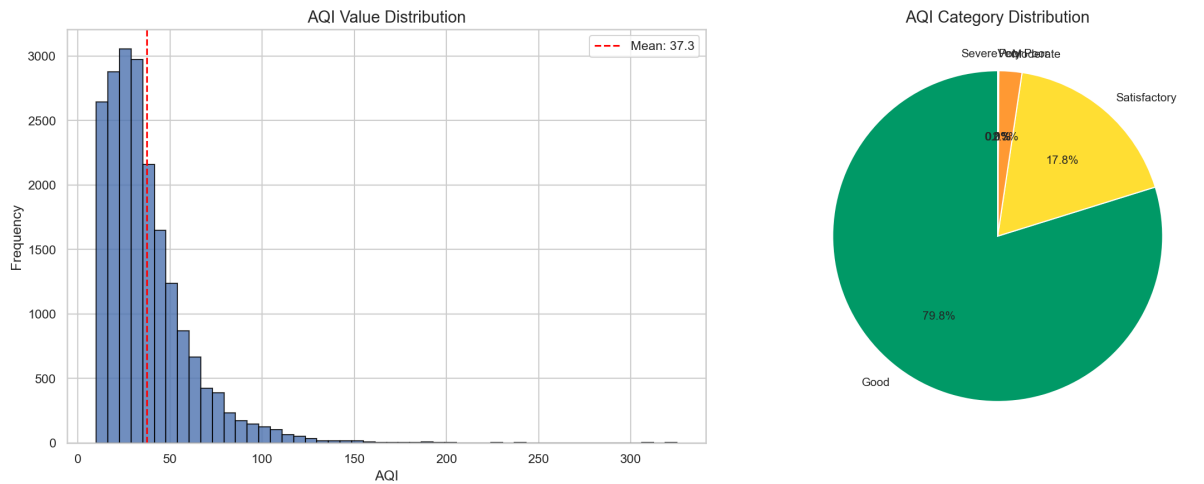


Fig 3.1: AQI Distribution across all locations

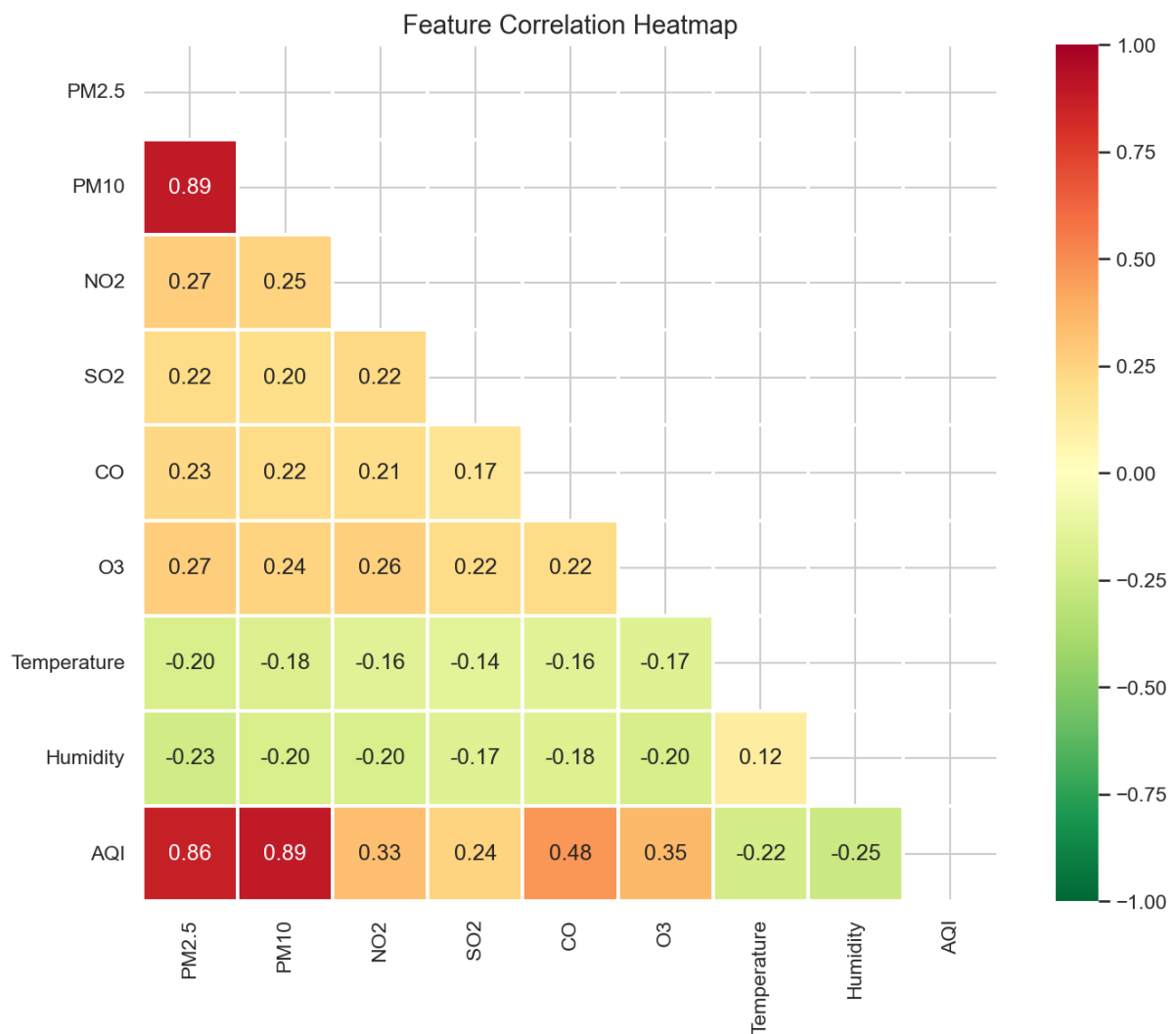


Fig 3.2: Correlation Heatmap of pollutant parameters

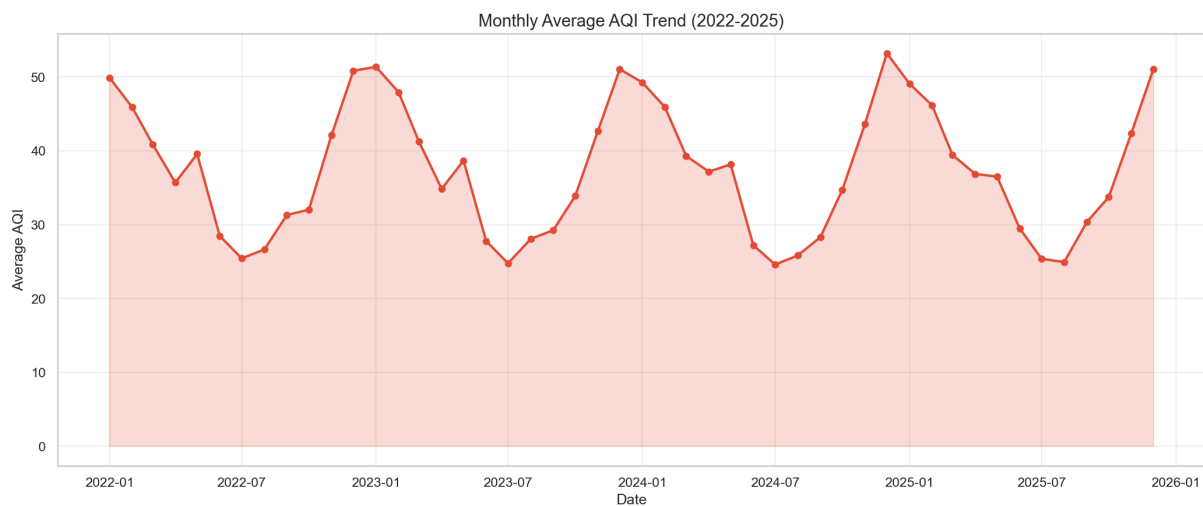


Fig 3.3: Monthly AQI Trend showing seasonal variation

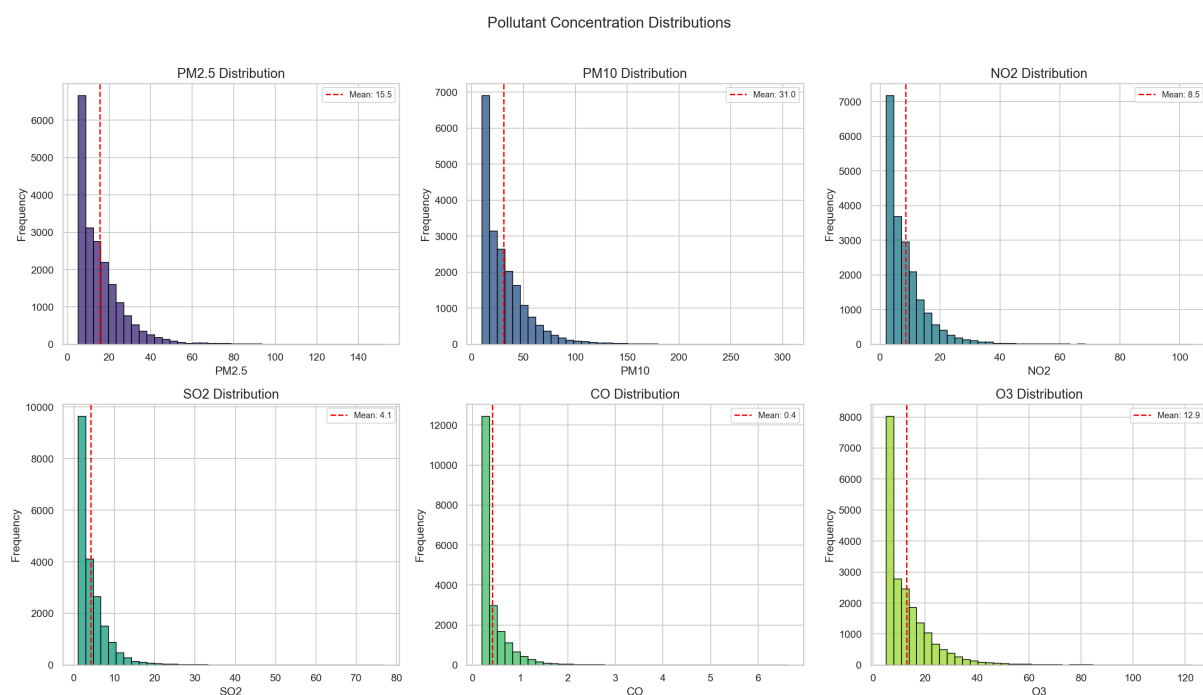


Fig 3.4: Distribution of individual pollutant concentrations

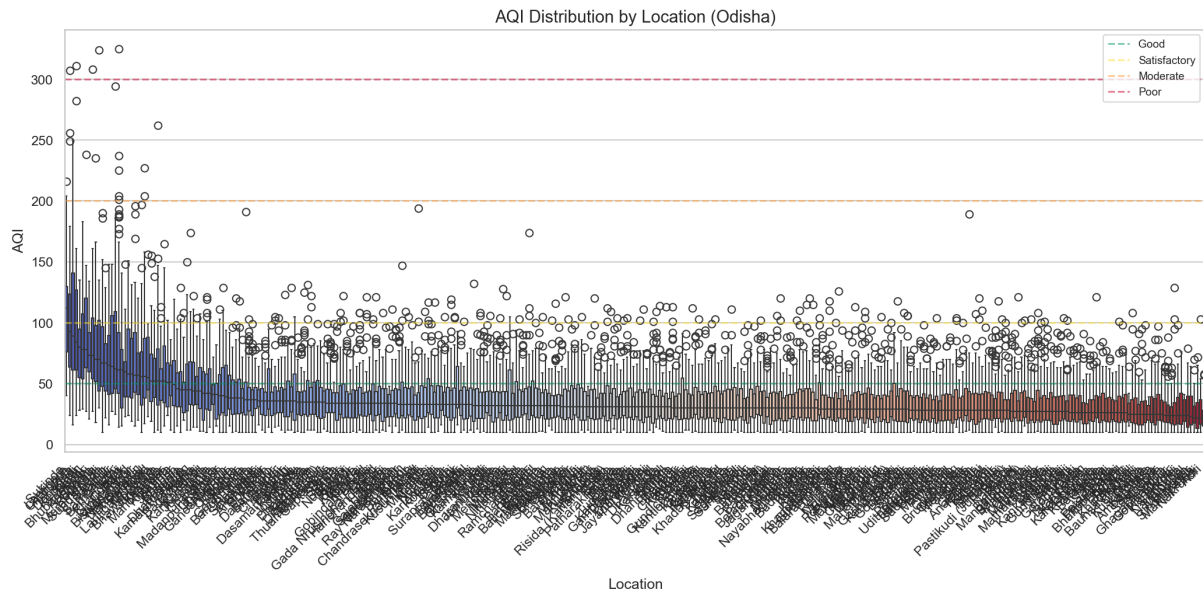


Fig 3.5: Location-wise AQI comparison

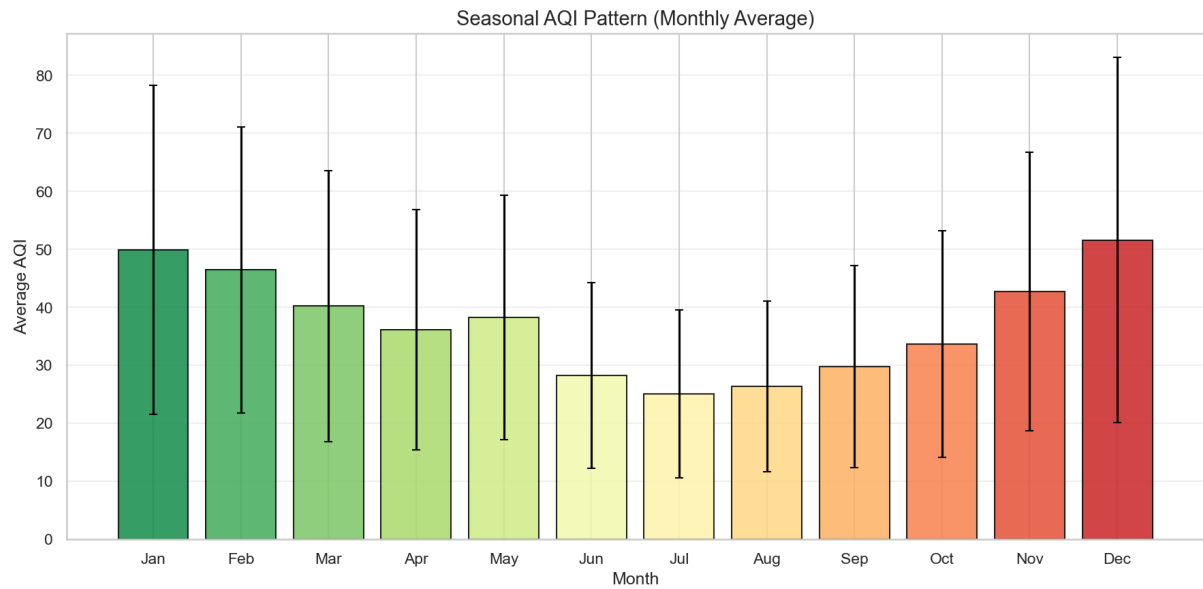


Fig 3.6: Seasonal AQI patterns

4. Model Training & Architecture

The system uses XGBoost (Extreme Gradient Boosting) as the primary prediction model. XGBoost is an optimized gradient boosting algorithm known for its speed and performance in structured/tabular data problems.

Hyperparameter tuning was performed using GridSearchCV with 3-fold cross-validation over the following parameter grid:

Hyperparameter	Search Space	Best Value
n_estimators	[200, 300]	300
max_depth	[6, 8, 10]	8
learning_rate	[0.05, 0.1]	0.1
subsample	[0.8, 0.9]	0.9
colsample_bytree	[0.8, 0.9]	0.9

4.1 Feature Engineering

- Temporal features extracted: Month, Year, DayOfYear
- Categorical encoding: Location and District via LabelEncoder
- 13 total features used for prediction
- Train/Test split: 80/20 with random_state=42

5. Model Evaluation Results

5.1 Regression Metrics

Metric	Value	Interpretation
R-squared (R2)	0.9868	Model explains 98.68% of variance
MAE	0.85	Average error of ~1 AQI point
RMSE	2.67	Root mean squared error

5.2 Classification Metrics (AQI Category)

Metric	Value
Accuracy	98.83%
Precision (weighted)	0.9879
Recall (weighted)	0.9882
F1-Score (weighted)	0.9881

5.3 Cross-Validation (5-Fold)

Fold	1	2	3	4	5	Mean
R2	0.9867	0.9814	0.9723	0.9835	0.9836	0.9815

Mean CV R2 = 0.9815 +/- 0.0049, indicating excellent generalization with low variance.

Actual vs Predicted AQI

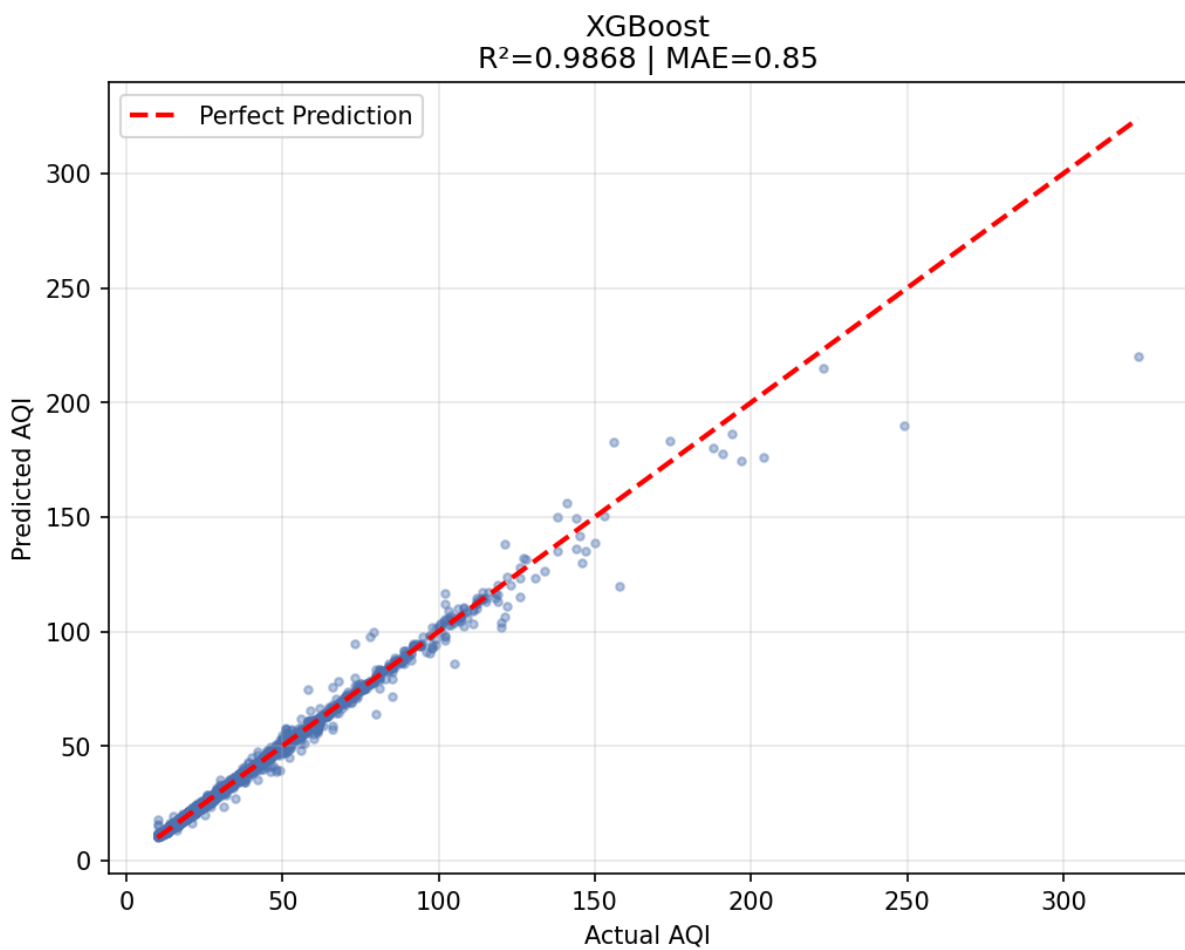


Fig 5.1: Actual vs Predicted AQI

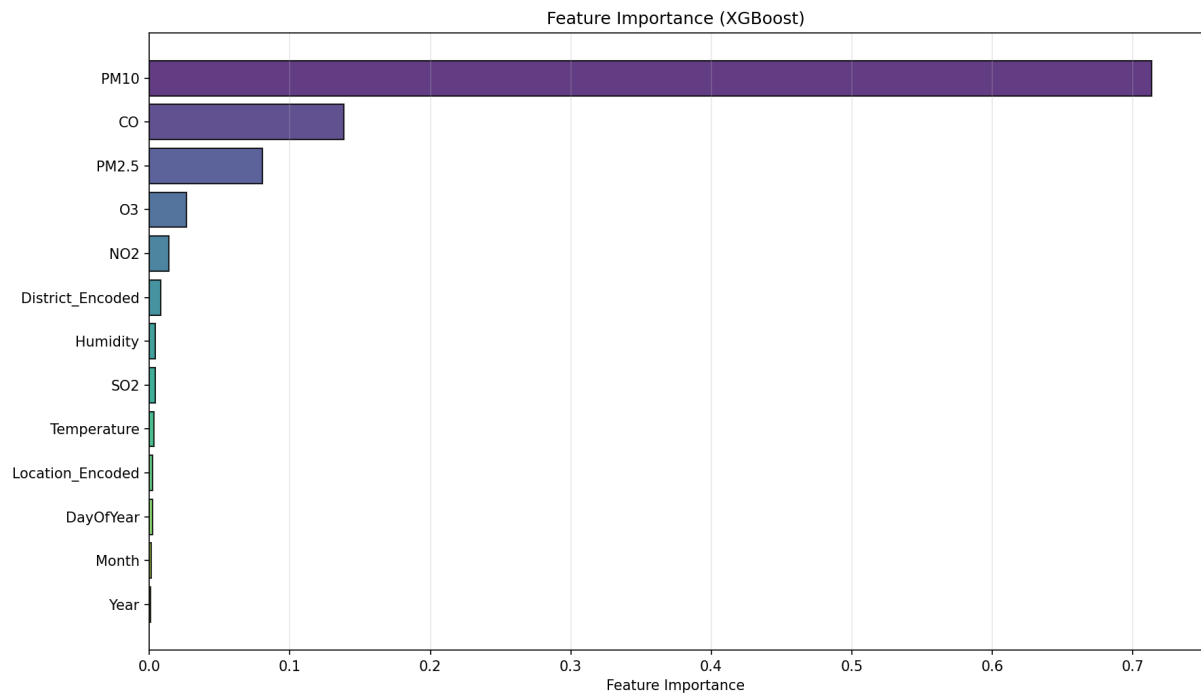


Fig 5.2: Feature Importance (XGBoost)

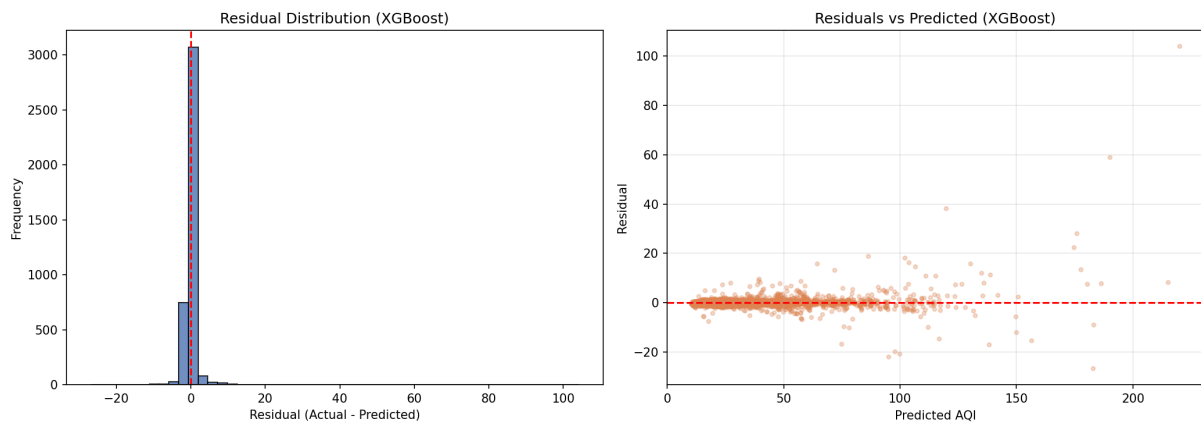


Fig 5.3: Residual Analysis

6. API & Deployment

A RESTful API was built using FastAPI to serve real-time AQI predictions. The API loads the trained XGBoost model at startup and exposes endpoints for prediction and location data retrieval.

6.1 API Endpoints

Method	Endpoint	Description
GET	/locations	Returns list of districts and their locations
POST	/predict/auto	Auto-predicts AQI (fetches live weather data)
POST	/predict/manual	Predicts AQI with user-provided parameters
GET	/health	API health check

6.2 Prediction Modes

- AUTO mode: Fetches live pollutant & weather data from Open-Meteo API, then predicts
- MANUAL mode: User provides all 8 parameters, system predicts AQI
- CLI Interactive: Step-by-step district -> location -> prediction flow
- CLI Batch: Process a CSV file with multiple predictions at once

7. Technology Stack

Component	Technology	Purpose
ML Model	XGBoost	AQI regression prediction
Preprocessing	scikit-learn	Encoding, splitting, evaluation
Data	pandas, numpy	Data manipulation & feature engineering
Visualization	matplotlib, seaborn	EDA & evaluation charts
API	FastAPI + Uvicorn	REST API for predictions
Serialization	joblib	Model persistence (.pkl files)
Live Data	Open-Meteo API	Real-time pollutant & weather data

8. Conclusion

The AQI Prediction System successfully demonstrates the application of machine learning for environmental monitoring in Odisha. Key achievements include:

1. R-squared of 0.9868 - explaining 98.68% of AQI variance
2. Classification accuracy of 98.83% for AQI category prediction
3. Robust cross-validation with mean R2 of 0.9815
4. Coverage of 400+ locations across 9 districts of Odisha
5. Real-time prediction capability via FastAPI
6. Live data integration with Open-Meteo weather and air quality APIs

The system provides a practical tool for monitoring and predicting air quality, enabling proactive measures for public health protection across Odisha.